

OSSP (Operating Systems And Systems Programming) - 25CS2104E

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Sec : 9

week – 1

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
int main()
{
    char command[100];
    int pid;
    printf("Enter a Linux command: ");
    scanf("%s", command);
    pid = fork();
    if (pid < 0)
    {
        printf("Fork failed\n");
        return 1;
    }
}
```

```

if (pid == 0
    printf("Child Process PID: %d\n", getpid())
    printf("Parent Process PID: %d\n", getppid());
    execlp(command, command, (char *)NULL);
    perror("Command execution failed");
    exit(1);
}
else
{
    // Parent process
    printf("Parent Process PID: %d\n", getpid());
    printf("Child Process PID: %d\n", pid)
    wait(NULL);

    printf("Child process completed.\n");
}
return 0;
}

```

Output :

```

pothubhavana@DESKTOP-16SJJUA:~$ nano week1.c
pothubhavana@DESKTOP-16SJJUA:~$ gcc week1.c
pothubhavana@DESKTOP-16SJJUA:~$ ./a.out
Enter a Linux command: OS sec-9
Parent Process PID: 541
Child Process PID: 580
Child Process PID: 580
Parent Process PID: 541
Command execution failed: No such file or directory
Child process completed.

```